6. Comparison: Layer-Level Editing (ROME)

Implementation on Pythia-6.9B

Mixed-precision ROME editor, rank-one updates on MLP weights at Layer 28. Tested on subsets of SET-POLY (N=30) and SET-MONO (N=20).

Group	Avg. Drop (P_B)	Conclusion
SET-POLY (High)	**65.75%**	Severe side effects.
SET-MONO (Low)	**21.00%**	Moderate (baseline noise).

Table 1: ROME results. ROME suffers in high-overlap but is much more robust in low-overlap regimes compared to FiNE.

Neuron-Level (FiNE) vs. Layer-Level (ROME)

- **High overlap hurts both:** Mechanistic entanglement (SET-POLY) leads to severe locality failures for both methods.
- **Layer-level is more forgiving in low overlap:** ROME shows much better locality in SET-MONO (**21%** drop) compared to FiNE's aggressive behavior (**77%** drop).
- FiNE's neuron-level patches appear **sharper**, driving probabilities closer to zero; ROME's distributed rank-one updates are less destructive when overlap is low.

7. Conclusion & Future Work

Main Takeaway: Polysemantic neuron overlap is strongly associated with poor locality, but neuron-level control is *not* automatically more local than layer-level editing.

- Strongly polysemantic circuits create genuine locality constraints for *both* editor types.
- Designing robust, well-regularized neuron-level updates to avoid "sharpness" remains a challenge.

Future Work: Full hyperparameter sweeps for FiNE; explore multi-layer patches; expand benchmark to diverse domains (e.g., reasoning).